

ActInf Textbook Group ~ Cohort 6 ~ Session 6 (Chapter 4, Part 2) ~ 3/25/20

TextbookGroup/ParrPezzuloFriston2022/Cohort_6 | Cohort 6 ~ Session 6 (Chapter 4, Part 2) ~ 3/25/2024 | 1:09:04

all right greetings thank you everyone
for joining we're in our second
discussion on chapter
four let's just jump to it um anyone can
write a question in the chat or raise
their hand or just like unmute and go
for it and it could be like anything you
remember from the chapter or like a part
you liked or we're curious about or just
any any question that you want to jump
in with
like any any section that people prefer
just un unmute just any page any section
oh Aon go for it so all um I I was
really scratching my head trying to
compare figure 44 with figure
46 um there's like
a very large amount of overlap between
the two sets of diagrams but there's
like some very uh so I feel like there's
like a clear parallel setup um there's
some edges that are different between
the two and I was like trying to
understand like what do those changes
mean um and like right these were yeah
uh and in both cases I think especially
the left hand side diagram is like
virtually
identical um between 44 and 46 I think I
have the numbering that right um but uh
yeah I I don't know if that's like too
much to fit into a question but I was I
I thought the difference there must be
meaningful but I I didn't fully I
couldn't make sense of

it it's a great question like for sure
on first pass it's like two cities like
maybe there's some differences and which
roads and what's connected to what but
at a first pass like they are both very
very similar um
so a few things are being shown here um
to to kind of give context on this qu
question let's pull back to figure 43 so
here we see this kind of like Rosetta
Stone figure with a basian graph for the
discrete time model and a basian graph
for the continuous time model on the
bottom they're kind of being like laid
out like this to emphasize their
structural similarity even though
they're treating time differently which
we can return to but the top is treating
variables in terms of discrete sequences
of events whereas The Continuous time is
treating it in terms of higher and
higher derivatives of a given um
variable X the important thing though
that makes these similar other than
their structure is that in a base graph
you have edges are variables and then
the uh I'm sorry nodes are variables and
the edges are causal connections okay so
now we get to aasian message passing
scheme so one of the really cool things
about AAS graph is because the edges
represent the sparsity of causal
Connections in the model you only need
to consider edges that are local to a
node so it's like the traffic at a node
is strictly and entirely defined by the
edges leading in um so when constructing
a given basian model like you specify

the kind of causal architecture of the world and then messages are passed between the variables so this is going to be a slightly different base graph than the figure

4.3 but it's going to emphasize how certain kinds of architectures pass certain messages to each other and then how that enables different kinds of computations to happen like there's a different variable here so like right off the bat even though we see some variables that were in figure 4.3 like π and S there's also different variables like the u ϵ and the IT u okay so two things are being highlighted in this figure

4.4 on the left is going to be about hierarchy and

nesting on the right is going to be about unfolding Dynamics through time so the motif on the left that describes the nesting is going to be like each unit of four is kind of like a level in a nested hierarchical model and basically what's happening is the E is calculating an error like ϵ for error and then those errors are being passed up to a higher order in the u graphs so that is like the kind of classical predictive processing architecture we have strictly stacked levels and then what's happening within each level is it's doing a differencing and then passing the errors up that's the hierarchical motif The dynamical Motif is going through time so this one actually has

more similarity to the discrete time in
4.3 and here it's kind of like a
predictive processing but emphasizing
the dynamic
here's the sequence of observations
coming in current time step the future
time step past time step and then each
of those observations are kind of
meeting in the middle and calculating an
Epsilon based upon the Collision of the
incoming observation and then the hidden
State prediction so like the Epsilon
would be zero if the hidden State
prediction was exactly what the
observation was and then those are
getting passed on

up I'm

um

okay then

4.6 here the only difference is that the
again the very similar shapes with like
the motif of four and the passing up but
now we're in the continuous time setting
so that's why there's um more variables
like we see in the continuous time

World um

and now rather than
representing the past present and future
time

step just like the top here past present
and future time step in

4.6 now they are

representing different orders of the
generalized ordinates of motion so the
first the second and the third
derivatives so 4.4 and 4.6 are kind of
like transposed from figure 4.3 the top
and the bottom but now it's representing

each of those halves in terms of the message passing that it reflects computationally so what's cool about it is it shows again a deeper similarity between these two kinds of models what's kind of challenging about it is it's introducing new notation and it's not exactly the same model as before

any thoughts on that or more more on that question or it could be a different question that anyone has

we can go to any any part of four or like any any can just anyone can just unmute and go for it or write in the chat yes

TCC

it's kind of like Parallax in Vision like chapter 4 is presenting the generative models of active inference so in terms of where we are in the textbook like chapter one introduced it chapter one two and three approached active inference from the low road and high road and then chapter four is like actually the kernel of active inference and then the discrete and the continuous time are going to be unpacked more in chapter seven for the discrete and chapter eight for the continuous time however here they're being really like tightly juxtaposed and interwoven again that speaks to a really deep point which is that time can be handled either in discrete or continuous fashion in active inference that's cool

deep Point what's challenging is that even when they're discussing similar kind of cognitive functionalities the notation is very different between the two forms it would be confusing in a different way if the exact same notation was used because it wouldn't be clear whether we were talking about the discrete or the continuous time however this is kind of like a second set of vocabulary but we couldn't use exactly the same because some of the variables are slightly different but like using Y for observations in the continuous time and then using uh o for observations in discrete time that's just how they do it in this chapter so it's kind of like they're introducing both time treatments and emphasizing again and again and again that that both of these are like two flavors and then they are going to revisit it in the dedicated chapter seven and chapter 8 but it's kind of like a double barreled chapter because they're introducing generative model not in terms of like well here's a simpler generative model now we're going to go to more complex they're introducing like a dual stream like the two simplest forms but anyone can just unmute and go for it or we can look at the chapter 4 questions or whatever people prefer okay if no one else has I'll go again I I'm confused also about the the structure of um uh policy variables like

okay so like a in a hidden State an
observed State whatever like I can like
sort of acknowledge those are sort of
like things that exist in like some not
fully specified space um policy
especially because I think it's like
trying to call back to reinforcement
learning literature like I feel like in
some what's what's like the type of that
like if you were to describe like the
space in which a policy either the
discret version the π or the uh
continuous version The V like what that
lives in uh or if you were to like
implement it in a program like what what
the type associated with that is like
what's it shape great question
 Π is a list of all policies so it's
always a vector and it always sums up to
one and that's really the critical move
that allows us to talk about the
likelihood of a policy that's why habit
is the prior on policy and then expected
free energy updates our prior on policy
into a posterior because the shape of
that variable is a list that sums to one
so it's a probability
distribution so affordances are the
actions that can be taken in a moment so
if we're like in a grid world the
affordances will be like up down left
right stay so it'll be like
five then if we're only a single time
Horizon a single time step agents
affordances are the same as policy space
because we're only selecting the next
action so say we have five options if we
had an even prior across those five it'd

be like 0.2 point2 point2 to and then
that could be sampled from or updated
into a policy
posterior if we're talking about an
agent that does planning then the shape
of P_i is affordances raised to the power
of the time steps so if we have five
affordances and then we're doing two
time steps then P_i is 25 long it's up up
up down up right and so it's all of the
two-way combinations so that's the kind
of
exponential explosion with policy
planning which we can come back to but
at a first pass the important things
about P_i are that it's a list it sums to
one which is what makes it a probability
distribution and it's a distribution
over policies that the agent is
selecting
from just to clarify or check my
understanding so um then whenever we see
like uh V with the Tilda representing a
trajectory over policies like that is
maybe should be regarded as something
that's like because as exponential and
like a unbounded number of steps is like
not something that we should consider
like concretely representable in a
program interesting so the The
Continuous time setting doesn't do
explicit
planning it does so the the PDP at the
top here this is where we have the kind
of classical computer science
exponential tree search like more and
more discrete combinations you know if
you had five times steps deep you'd have

five to the fifth discrete options it's kind of like a chess algorithm um in the continuous time setting planning is not explicit like let's just say that we had some wild function that we were trying to approximate when doing this kind of a Taylor series approximation like evaluate the function at zero and then take the first derivative so let's just say we stopped there we're just going to evaluate the function and then look at the local slope that already expands to everywhere that doesn't mean it's going to be accurate everywhere but basically the Taylor series does expand no matter how many levels deep you go it already implicitly includes basically infinite future planning but for explicit planning in a tree like setting that's where the discrete time comes into play in um in this page um Thomas P's discussion so early implementations were in The Continuous State space because this has the most similarity with like Bayesian filtering approaches then people started wondering well how do we get sequential Dynamics then they moved into this kind of coupled dynamical systems space but then that was only getting so far because there wasn't an explicit representation of planning then they moved into the discrete planning

environments those branches kind of came back together by nesting The Continuous Dynamics at the lower level with the discrete Dynamics at a higher level so you could do a lot of work in the continuous space and then also have that like wrapped by a discrete planning step that's like a little bit about how the timeline worked but just to summarize in the discrete setting this is the most like classical planning in the continuous setting you don't have planning as such it's kind of like you have a car going around a turn in the discrete world Step uh discrete time world it's trying to estimate all the branching paths it could take in the continuous time world the driver's like I'm turning 20 degrees to the left that's like the first derivative of my turning angle and then I'm just I'm holding to that and then that curve continues so it's not explicitly saying at seven times t now here's where I expect to be so it's the strength and the weakness of the discrete time the weakness is that there's an exponential explosion of the tree just like a chess algorithm but the strength is that you can actually talk about the difference between up up up up up down and up up up up up right which you couldn't really get except by going super deep in the space so they both have these kind of like strengths and weaknesses and then that is resolved or addressed by

building hierarchical nested models that have components that have continuous treatment of time and components with discrete

good call um let's see

in page

73 yes someone else has already added

this but yes good call there's several

typos in the

book

we can go to any of the written

questions or like just any any remark

about how anything that we've gotten up

to or just like anything anything active

related we can totally go for it because

chapter four is just it's kind of like

the heart of active inference

um albeit described in a very um

notational

way David or anyone go for

it uh yeah I was just wondering uh if

there's any relationship to the partial

gamma function in regards to how the

branching um seems to work and fit

backwards like if you're mapping back up

through the stack of the decision tree

that Maps back to the initial I just

didn't see any of the the math there in

the chapter but I've seen some

correlation in some of the um the math

that I've been working on

recently um so if you go further down uh

there's a section on

branching

and the relation yeah so Behavior near

Branch point and the relation between

the branches so I was just wondering if

um if that fits in or if my intuition on

the the correlation is incorrect
there it's an interesting connection I
don't know about like this analytic
family but a few thoughts are inactive
inference we're treating every moments
as a branch point so that is the
control decision slash question is the
branch point is defined by the
affordances that can be taken at that
point and so there's some um you know
there's some kayaking down the river
where no matter which way you Branch
it's like you're going to get kind of
brought back in so like the long-term
consequences don't matter but then
there's other moments where to to go a
little bit one way or the other is going
to
diverge so that is kind of like you have
the agents on the kayak and they have
the behavioral Branch point but of
course just knowing that they can tilt
left or right doesn't tell you about
like the river which is kind of like the
actual process
unfolding so I'm not sure exactly how
that would connect but action is a
branch
point
another another possible connection
there is like in this chapter we're
seeing like the basic kernel like the
minimal essential model that describes
um either you know in either of these
time treatments but then there's all
kinds of other cognitive functionalities
like what's known a sophisticated Act of
inference is where in the moment you're

not just planning um or or trying to
estimate like what will I be doing and
what will I see but you can
also project out what will I be thinking
then about the past so again that
explodes the state space even further
but it allows for like sort of
prospective
retrospective but that that kind of
phenomena is not
here but it's like a modification on
this base
model yeah I guess I was seeing it more
in terms of um the primes and the double
primes on the uh on the bottom set and
maybe that was just my miscorrelation of
just misreading the um terminology as
it's as it's represented for the uh the
lower States so you see I um y y Prime
and then y double Prime
there that's the first derivative and
the second der
right but in order to move back like if
you were if you were collapsing back
into the initial kernel stage wouldn't
it have to still
step through th those premises in that
order or would it all collapse at the
same exact
time again there's a lot to this
question I mean what I'll say is the
role that three plays here the B Matrix
kind of conventionally this is a
transition operator that's describing
how things change from time step to time
time step whereas the three operator in
the continuous setting is describing as
we go to the

right the derivative and as we go to the left the integral so like you said you have to walk back through the premises um step by step from 10 times steps forward you can go 10 9 8 let's just say but you can't skip from 10 to eight at the very least you would need a different transition operator to apply backwards similarly um you might be able to go from X'' take the integral and then take the integral again but you couldn't take a double integral in one jump that wouldn't be the same as these integrals

okay

but it's kind of like yeah there's other ways to take it like the computational irreducibility angle which is like there might be a shortcut to just apply the double integral in one step but then there's also probably equations that you like can't apply a double integral and in a way by doing a double integral in one step you've just made a single integral you know or by going two time steps backwards here you've just taken one of a different kind of Step okay yeah because I ran into something recently that resolved into a five integral and that's where I got confused as to how that was even possible and I'm still trying to work through the math a little bit to

understand but uh that's kind of out of the scope so I don't want to take too much time out of that if you're looking at something

that's that's a really long sequence you might be able to have statistical power to talk about like the fifth derivative but you could simulate trajectories and then just plot the first derivative plot the second one plot and so on and then you'll just find that maybe there's interesting dynamics of higher orders or maybe not and then the empirical way that you could ask like how important are they would be okay so we make a real simulation that five um five actual derivatives of influence and then we make a subset models that have all five that should be able to recover 100% of the variance and then you test the one with four three two one and there's situations where even if there's really five the first derivative covers 99% of it or there might be ones where the higher derivatives do matter okay thanks

here just just again just to Connected here's just evaluating the point so to say you know this is this is like the true sea levels for the next 24 hours or this the real temperature in the room so it's like if you can only have one number the best thing you can do is just say what is the temperature right now and that's this kind of first approximation and then if you were to have a second number you could say okay what's the temperature in the room and then what's the the

slope and then if you had another number
temperature right now and the slope and
how fast it's changing and so that
process
is the Taylor series
expansion so that's the kind of way that
they talk about it
in approximating functions because you
can keep on adding more and more
approximating
terms but it can get
unwieldy and you still get and yeah
there's other things to say
but this but I guess I'll just say this
dash line does go off to Infinity like
you could evaluate that dash line at
this point but it' be like way way way
down and then the one of the issues with
using the um Taylor series is you don't
know how good or how bad the dash line
is except by then doing the laborous um
work of checking against the actual
function so like here it's like it
actually is doing really well but it's
not telling you it's doing well you
would just find out by checking but then
over here it's going to be doing
absurdly bad but it's not like it's
going to throw up a yellow flag and tell
you I guess my question in terms of uh
that that dotted line trending to
Infinity is is that assumptive or is
that you know provable in the sense that
we just can't see past a certain State
because it's you know a hidden state or
is it calculable past that hidden state
that is going to be
infinite the dash line is very well

behaved because it's a polynomial okay it is this

polynomial here's the first order polynomial just a

constant here's the second order it's a linear $y = MX + b$ basically $MX + b$

intercept now we're in the quadratic polynomial setting so it has all the kind of simple behavior of

polynomial which is what makes it a well behaved approximator for arbitrarily unruly

functions but it doesn't tell you how good it

is

okay also just note like on the pragmatic side pmdp the python active inference tool kit is only in discrete time at this point right I'm trying to translate some of it over too because I'm starting to work a little bit with kiss kit or I'm trying to uh to understand to try and you know use some of the functions there with Quantum Gates but that's a little bit beyond um to try and utilize that as a tool so I'm trying to put it together a little bit and then I just got a little bit confused over the last few days putting some things together so um I'll keep looking at it appreciate

oh all

good okay anyone else just unmute and go for it or feel free to type a question or we can look at the the upvoted questions here or any question that was written in the um chapter six or cohort

six

questions

check okay n

specific

I think it's worth to kind of pull back

a step and look at figure

4.2 figure 4.2 is really helpful because

it kind of builds intuition for what

these base graphs are in terms of like

here's a situation on top left with one

hidden State and one

observable like there's a real

temperature in the room and then there's

a thermometer here's the situation with

one hidden State and two

observables here's a situation with two

contributing factors and one observ

like two things that contribute to a

disease prevalence and here's a chain of

causation so these are like these are

the kinds of things people say

like there was traffic because it was a

sports game and a Friday night it's like

okay that's this

situation or it was a Friday night so

there was a sports game so there was

traffic but those are are the kinds of

things that people talk about when they

use language to describe causal

sequences in the world and then these

variables can be like anything like the

observable could be any modality of

observation and then the hidden States

can be any latent causal Factor

including things that don't like

actually exist in the world like without

arguing too much about whether

temperature actually exists in the world

you can have latent causes that are just totally proposed for better and for worse but this maps onto like a huge fraction of the kind of statistical inferences that are done and this also is kind of like if we had like a mind map and we were just like drawing out causal factors for what we cared about like this is kind of intuitively how you would do it like if one thing was causing multiple outcomes you draw it like this if there's multiple things contributing to cause an observation you could draw it like that so this is kind of like where we touch with more intuitive notions of cause and then that is going to get deployed in this setting where we're talking about the causal relationships between observations hidden States and actions those are the big players here and that's being unfolded through time but those are the things that we want to connect in our unified model of perception cognition and action is to a first approximation perception cognition about hidden States and action that's why those are like the major nodes in the white circles has someone tried to implement this for like motor control yes um motor control has been approached

from the human motor control perspective
like handwriting and movement of the arm
and things like that and also the
robotic motor control setting at the
kind of gross level like where to move
within the warehouse and at the finer
scale like robotic arms and things like
that Dan wasn't that related to what the
Hitachi group

covered I just thought that might be a
good video to point to
yeah good call thanks that was um
guestroom

75 so here was a
robotics in silico so it was not
physical robots it was a simulation of
robots but but this is um a group that's
working with physical robots
and then they
um the visualizations that they show in
the in the Stream were were pretty
interesting and provocative I think in
the sense like it it

[Music]

um it shows how multi-agents and also
how social
norms come into
play so here we have basically the the
red agent is the active inference agent
and then it's going to be like coming on
the opposite direction of the
sidewalk here it's an altruistic other
so the green one has the goal of making
the red one go as little out of the way
as possible whereas the selfish green
one is trying to minimize its
path so those are kind of ways in which
people implement like even these kind of

social characteristics into to the
preference distributions and the action
Tendencies of different
agents and that's kind of the journey
that chapter six takes us on like which
so that's kind of why the book can be
read in many orders because as it's
written linearly the first five chapters
are like a ton of information that's the
epistemic component and then the second
half of the book starting with chapter
six is like well here's the recipe for
for building the model like what are we
modeling why are we doing it what form
should we use figure 4.3 so just the way
the textbook is written it kind of shows
us a lot of the tools in the workshop
and then we get the recipe in chapter
six but another way to do it and of
course if you go through many times then
it doesn't really matter as much where
you began but another way to to go about
it is start with the system of Interest
Like A system that you're familiar with
or curious about and then start to
reverse engineer like what are the
observations here like who's observing
what I mean I could explore even in this
Starbucks like who's observing what what
are the actions that different agents
are taking what preference distributions
do they have you know some people want
this level of sweetness or this um kind
of coffee
so that's like the kind of low road
approach is starting more with an actual
system and then building up to what
formalisms come into play rather than

like learning all the math
equations and then just hoping that
something like crystallizes out of
nothing thank you Cheryl I'm gonna I'm
gonna bring this in let's look at it
awesome yeah the the these are great
points um like well you know the the
brain is a system of systems so like if
we
up here this could be like this is like
our let's just say this is our portfolio
models I mean the brain entertains
multiple causal hypotheses about
events so it is a system of systems and
these are these are yeah good ahead I
guess I meant it's not a system of just
internal systems and that's where I get
um mixed up I mean it's the whole of it
the whole experience in the world we are
not just brains we're in ecosystems
we're in Social systems we're in you
know what I'm saying um
yeah it's like it's such a great and
essential Point that's why we focus so
much on the action perception Loop which
initially seems kind of esoteric but
then we start to see that like all
agents around us are involved in it and
then the only way to account for that
kind of like Dynamic
co-creation with the environment and the
agent is to think about the interface
every time step like if you're on a run
or if you're driving
you can't just focus on well just ignore
the perception for the next time steps
or just ignore the actions for the next
time steps like action and perception

are always happening the agent can't opt out of those happening and so it is a joint entanglement with the environment and that's like that's exactly what has to be kind of modeled to have a holistic model of even just a rat in a teamas correct correct so you're right I like it I like the systems in systems within systems model but we can only work with what we can work with that a time that does infer a certain amount of reductionism though as if if I understand it at this level I will understand it at the more at the larger level so that's a that's a an assumption that we bring to this yeah I again you know there there's no single order for going through active inference but figure 9.1 kind of captures that like here in the center is the top of figure 4.3 so this is the rat in the teamas that's the subject of our study here's the scientists in the lab so what we choose to present as stimuli are observations for the rat but they were choices by us and then the choices of the Rat become observations for us and so you could close your analysis here and say well it's just a rat in a laboratory or you could model the building around the laboratory and you could ask like how does that context matter and then you could ask well how confident are we that the the rat behavior in the lab is like it is in the

field there's laboratory setups that it probably transfers and there's ones where it doesn't but just making the model of the rap behavior in the lab does not guarantee transfer and and that that's why the system of systems is so important right okay I love that but I'm not in chapter nine metabasian inference oh my gosh I don't even know I'm gonna get out of chapter four okay so I'll wait I'll wait there but um I do I do have a question have you ever heard of Marilyn Monk who's also at UCL uh she's a molecular biologist she has a really interesting um theory on uh cause Consciousness yeah and and um she takes the systems all the way from Adam to Cosmos that's what that's what got me uh thinking about what we're doing here in this in this book but uh a very interesting model that's cool like definitely some people are going that kind of like multiscale like down to as small and fast as you can imagine like Quantum and then up on through the cosmological and then this seeing this part here that reminded me of of a really great book um evolution in four dimensions which is like there are genetic epigenetic and metabolic and cultural and symbolic inheritances and there situations where one explains all of the variation but there's situations where it explains none so it's not like you can just have

this in principle battle and say oh you
know epigenetics wins or symbolic
variation wins it's like the the field
has been
sketched and then the empirical question
is what matters where and how and then
what but those are questions to address
four specific systems in practice
they're not questions that can be
addressed you know dusted off once and
for all in principle yeah yeah yeah yeah
yeah yeah I mean all right yeah there's
hope for me there's hope for me then
there always

[Laughter]

is it's I mean it's it's fun and it's a
learning there's many things that that
connects because like we're talking
about the systems that we talk about
elsewhere so when we talk about
organisms we're we're we're talking
about the same we're not talking about
some other system and this is more like
a lens that we bring to
systems right a lens we bring the
systems another way to think about that
is like act active inference says less
so we can say
more but it's not like active inference
is g to you know it's not a slot machine
it's not going to deliver us like an
account of a psychiatric or a social
situation I mean the textbook is what it
is like
numbers are just there for people to
count things let's just say so these
models exist for us analyze cognitive
systems but if active inference already

had an answer about how cognitive systems are right that would that would curtail it from being used to explore diverse intelligences and open-endedness of creating new

things

yeah got

it and that's why like yeah thank you

that's why chapter four gives us the

like the generative

model it's like this is like the Linux

current or like the software system but

it's like but what do I do with a

computer it's like well chapter 5 here's

people using it to study

Nels but in chapter four we see it in

kind of its pure

form in the general or in the abstract

form which is like both sometimes very

frustrating or it feels like it's like a

mega infinite Learning Journey but all

of the connection from ACM to any system

we make that connection that connection

is not for us to

find

Dan could you uh go back really quick

for me to uh the chapter was I scroll

down I think it was five where you had

gone down to uh Consciousness related to

that um we introduced um the

differential between animal and uh

trees this quote here or plants yes

sorry

um so in terms of they have no brain so

they're called plants but in the in the

context of the tree itself isn't that

wouldn't that be considered

um in the same way that Cheryl was just

talking about different layers of the brain where the tree structure itself right has its own branches and nodes that are still connected in the same field that may grow around but again you're also deal dealing with different Dynamics where those systems will still interact given how close they are for Fitness of reaching the sun you know variably if another Leaf is over another leaf for example I totally agree like it's kind of a um it's the kind of quote that a mammal said like I think about this paper this is my PhD adviser and a colleague a framework for plant behavior and I mean now this is more of a common perspective but um there is some question whether the words to describe animal behavior such as the word Behavior or foraging can be applied to the activities of plant I mean is a root tip not foraging for nutrients in a niche oh it's slow okay but not to itself it's slow to a person running around but um so oh clant don't have Behavior they're just there but they do have behavior and then active inference helps us say yeah here's the affordances they have and the time skills over which they have them but you couldn't say like something you know doesn't have Behavior just because it doesn't have a kind of behavior right well

that go ahead go ahead sorry go ahead no

go ahead Char sorry

I I was just saying that um Behavior Uh
assumes

Consciousness and Oxford dictionary
defines it as the state of being aware
and responsive to one's surroundings so
if you like Marilyn monk she talks about
awareness is sensing so if we say
sensing and responsive to one
surrounding then we have to say plants
have a Consciousness according to Oxford
dictionary so what you're what you're
saying saying about the leaf and the
tree and the root

tip um brings the word Consciousness
here which is problematic for me because
it has no end I mean what doesn't sense
respond so I guess say even rocks do
that that's that's these are the mega
philosophical questions like does pan
cognitivism the idea like we approach
all things as cognitive systems or that
they truly are cognitive systems does
pan cognitivism imply pan
psychism is there something it's like to
be anything because all things are
engaged across the interface in sensing
and adapting even something that's inert
it's still is bounded by an interface
and there's still something on the other
side of the interface even if it's just
a cup of water but then is there
something it's like to be a cup of water
because it'd be different if the cup of
water weren't there so it's doing
something in relationship to its
Niche and so that's people looking for

continuums where it looks like there's sharp distinctions and looking for Sharp distinctions amidst the continua like it has to have a higher order awareness like it has to be aware that it's aware like our attention is not only about Target of our attention we're also aware that we're paying attention to something and this is where so many of the people bring in like meditation and the mindfulness connections and then some people use that to differentiate or distinguish human cognition from other types other people deploy like literally the same formalisms to argue for the continuity of human cognition and then this has been oh yeah go ahead um has there been discussion around like how much our sensors um contribute to our cognitive function say I mean just just as a mind experiment you if you took a human brain and you put into a dog's body would we start thinking like a dog more like people have thought wild stuff well I mean just to come at that from a different perspective I was thinking about this the other day in terms of cybernetics right like if we get to a point where we're able to do a full transplant of you know a mind for example into a cybernetic system wouldn't that be the same argument you know would you have the same kind of organ rejection for example um but it

would would it

be and I guess that would be very
assumptive at this point but like would
the the false body reject the mind or
would the Mind reject the false body in
that

sense I was thinking of more from this
perspective of you know how necessary is
it for intelligence or like general
intelligence like you know do want if
want to get high level's intelligence do
we need a an embodiment system that
you know senses the
world um in in a complex way that you we
have you I think we have
like uh 100,000 like Sten receptors or
something our body and it's continuously
feeding into our body for many different
layers and you know maybe that's
something that that's crucial to you
know the feelings and emotions that we
have um you because there theories where
you know like our our stomach itself
contributes to our you know cognitive
function

so you know I wonder there's been
discussion there from you kist and or in
the active influence Community because
the um you

know this community seems to think more
about um I think the

holistic

um yeah just a bigger picture in terms
intelligence so I did my PhD at the
University of Virginia in the United
States and there was a little known
secret there of a guy who studied
near-death experiences reincarnation

from all over the world Ian Stevenson was his name uh they didn't even really talk about him until years and years decades later um he was a well-known secret but his point was that uh Consciousness can exist without a body and even without a brain and at that time it really got me yeah there he is uh it really got me to thinking about this whole idea of Truth um there is a scientific model of Truth in which of course this is done by humans but before there were scientists there were still people seeking truth and even in Act of inference they're called the mystics um and then I think in our day you know there there are poets who kind of intuitively or think deeply about something but it's still a search for truth so I think um for me it has been useful to acknowledge these different ways and perspectives and modalities and not think one is greater than the other that's a great Point there's so much to say it it reminds me of like there's the parable of the the blind people and the elephants and they're like touching different parts of the elephants okay now we could kind of modify that into one person can see one can hear one can smell one can talk so there's different modalities that have not just touching a different part of the body but they're actually getting like a different wavelength they're getting totally different modalities of

the
elephants um that's that's one
interesting component and then one other
point is like um so earlier asked like
what is the role of the sensation and
perception in intelligence
there's a ton of answers but one answer
that like Adam Saffron and others have
pursued is um the kinds of relationships
that our sensory
apparatus require to
solve help us build architectures that
reflect the structure of our reality
like grid cells in the brain those
evolved because we were moving in spaces
with certain geometries
but then those became later utilized for
other kinds of arbitrary cognitive
navigation or like having hierarchically
nested perceptual
apparatus it arose to
do something that it was repurposed to
do later because that Motif is so
effective at the kind of statistical
challenge that it evolved to solve or
address and there's probably no end to
those
motifs
wow yeah it's super it's like chapter
four is so hard this is where it has to
take
us can I I know over time can can I ask
like a very fine Grand question to zoom
all the way back in yeah yeah um so I I
kept getting confused in the text by
some of the conventions in the math um
are bold faces and our serifs versus S
serifs meant to mean anything so if you

look at equation 4.11 for example we have caps SF f equals bold π times bold SF F and I I I looked around I couldn't figure out like what if anything are those like changes in the specifics of how things are rendered meant to indicate

I'm totally with you that's one reason why like having the um having the written equation

form I feel like is really vital because even even experts will not be able to converge upon this kind of a

description without looking at the same page there's just too many ways to read even the equations and that's even before you get into like bold and italics and like font and stuff like um it looks like bold is only mentioned one time in the supplement or appendix B referring to it as being like a sufficient statistic but I don't think that's how it's being used above I feel like bold is being used to describe whether it's a vector or not but

I I I don't know and like faced with that daunting of trying to sift through where they or if they Define it for like typographical phenomena that I don't even necessarily know the name of that's why I'm very excited for ontology driven natural language descriptions of the equations and then we can render them to look however we want so we could render them with a full English word or we could render it with

a full non-english word or we could use
a python

variable because I I don't even if you
could dig to the bottom and find out if
they defined how they use

notation other papers are going to do
different notation so it'd be kind of
like a a local

victory yeah at this point I would be
satis or I would be helped by local

Victory there's like a couple of pages
where they like seem to use this
distinction a lot and I was like not

following their choices at all um but
but yeah I appreciate that like a shared
resource like helps create some clarity
um but uh

yeah okay equation

411 uh yeah is an example and then like
above and below this there's a few
there's like a number of others that are
uh I think in to use the same

distinction and I I have to assume that
the authors had some purpose in
mind so I don't think it's just like

Matrix versus Vector versus
scalar

and the question is what does the Bold
typing versus the italics mean what does
what does the bold mean and what does
the serif versus un serif
mean

okay and if if no one knows I guess no
one knows and that that's the state of
the world but yeah I was I was what do
you mean by S you mean this F versus
this F right so they are distinguished
by by the Bold feature but like other

things are also distinguished by the
Bold feature but not by the SF versus SF
change okay it seems like we always have
a co-occurrence of
bold and

straight and italics is never found as
bold

so I'm tempted to say that bold and
straight means sufficient statistics
of whereas the italics is referring to
the underlying variable as it is but we
can like literally just write the
question down in one of the rows and
then whatever we can't resolve we can
email friston at all okay fair enough
yeah

yeah yeah I

mean I dream of the day when we can
Mouse

over and have multiple modalities of
what these are and and like have someone
says I want to Mouse over and see
examples someone else says I want to
Mouse over and see like the programming
variable I I don't know if there's
another way to make like a fixed
artifact that that

could speak to so many people in so many
different

situations okay all right

well interesting times question uh yeah

yeah um I then coding exercises that I
can go for to um understand these um
equations

better yes i' go go over the code page

here's several dozen um code uh examples

like when you see and then also I would

highly recommend the PDP

documentation because like when it's shown as the base graph and it's like it seems like again it seems like it's going to be like this math and analytical thing but when you go to active inference from scratch you you define ABCD and then you have all of the steps all of the utility functions are are defined and then um you have the active inference

Loop and then once you've gotten to here here's active inference Loop A B C D and the environment and the time

Horizon so if you've um run up at the equations Andor you want to try a little bit more of like a bottomup computer science or computer engineering approach this can be really clarifying but those are very useful with

PDP um but any of the code examples will be fun or useful however pmdp and RX and fur H um have the best

documentation so and we're here to make the the material more pedagogical so if there's like a example that you're excited about like Lally let's have the authors do a live stream or let's make the code better annotated or documented or whatever we need to

do um do they have any examples where they're doing um active influence on just some

basic um basic

predictions like I don't know tempature or yes like um

the uh RX

infer

documentation has has like a range of
examples um ranging from

like don't worry about something being

too simple truly like even the

examples um

of coin flipping are a great place to

start but this is a ton of information

about all kinds of examples and

then usually the people who are making

the packages are focused on making the

package if not like working three jobs

so if something's missing I mean change

the world or change your mind so if you

think there could be a cool

example that's what we're doing here so

I hope that we can make it if it's not

already

there yeah wow that was chapter

4 the next two weeks we'll come back for

chapter five and look at more examples

in the Maman nervous

system and then we'll have just a final

meeting kind of catch

up hopefully get people's feedback and

connect on projects and just see where

things are at in The Institute the

ecosystem so thank you all see you next

time thank you bye bye thanks everyone

thank you